

CAPE DECISION, AK, USA

WMO: 703880

Lat: 56.000N

Lon: 134.133W

Elev: 49

StdP: 14.67

Time Zone: -9.00 (NAK)

Period: 86-19

WBAN: 25315

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.0	24.8	3.6	6.6	23.4	8.0	8.3	26.1	50.9	40.4	46.6	40.6	13.3	60	0.810

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	63.7	56.1	59.8	54.5	57.9	54.0	58.3	64.9	56.2	58.7	55.1	57.2	6.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.4	65.5	57.7	54.5	63.2	56.8	53.6	61.1	56.0	25.2	63.9	23.9	58.9	23.2	57.2	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
44.6	39.1	33.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	38.9	38.1	39.1	42.9	47.4	51.7	52.9	54.1	52.0	47.5	42.1	39.2
	DBStd	7.51	6.28	5.95	4.80	3.13	3.14	7.73	3.27	3.05	3.62	3.32	4.48	4.82
	HDD50	2051	345	333	338	214	92	33	11	7	20	87	237	334
	HDD65	7126	810	753	803	663	547	421	374	337	391	542	686	799
	CDD50	425	0	0	0	1	11	84	102	136	80	10	1	0
	CDD65	22	0	0	0	0	0	22	0	0	0	0	0	0
	CDH74	358	0	0	0	0	0	358	0	0	0	0	0	0
	CDH80	231	0	0	0	0	0	231	0	0	0	0	0	0
	WSAvg	14.1	17.9	15.3	14.7	13.9	12.7	11.0	11.0	10.5	11.8	14.9	17.2	18.1
	PrecAvg	104.8	11.5	8.6	8.4	6.4	4.6	3.8	4.8	6.7	10.7	13.4	12.9	13.0
(15) Precipitation	PrecMax	142.7	21.4	17.8	15.8	9.8	11.5	7.0	8.9	14.6	19.3	20.9	23.1	28.2
	PrecMin	75.3	3.5	3.1	3.0	0.6	0.3	1.4	0.4	1.7	3.9	5.9	5.9	5.9
	PrecStd	17.0	3.6	3.9	3.3	2.3	2.6	1.4	2.0	3.0	3.7	3.4	4.1	5.4
	PrecStd	17.0	3.6	3.9	3.3	2.3	2.6	1.4	2.0	3.0	3.7	3.4	4.1	5.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.8	49.4	49.6	54.9	60.1	98.8	63.5	65.1	62.2	56.5	52.4	49.0
		MCWB	42.6	43.0	42.4	45.9	48.5	65.5	57.3	57.0	53.7	52.7	49.1	43.8
	2%	DB	48.3	47.4	47.5	51.0	56.2	88.5	60.1	60.9	59.1	54.3	50.5	47.3
		MCWB	44.6	43.0	41.9	44.2	48.1	61.8	55.3	56.0	53.8	51.4	48.0	44.2
	5%	DB	46.9	46.1	46.3	48.9	53.3	59.1	58.2	58.8	57.2	52.9	49.4	46.1
		MCWB	44.2	42.2	41.4	43.2	46.7	53.7	54.6	55.1	53.3	50.0	47.0	43.1
	10%	DB	45.3	44.9	45.0	47.3	51.4	55.7	56.6	57.3	56.0	51.7	48.0	45.0
		MCWB	42.8	41.5	40.6	42.3	46.7	51.9	53.8	54.4	52.7	48.6	45.3	42.2
	0.4%	WB	47.6	45.8	46.0	47.8	51.4	65.9	58.6	59.3	57.2	53.9	50.2	46.5
		MCDB	48.5	47.5	47.9	52.3	57.0	99.1	63.0	64.5	59.7	55.4	50.9	47.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.0	44.1	43.7	45.8	49.7	62.0	56.4	57.4	55.4	51.8	48.8	45.2
		MCDB	47.4	46.0	46.1	49.9	53.5	89.2	59.0	59.9	57.7	53.2	50.1	46.8
	5%	WB	44.3	43.1	42.5	44.3	48.4	54.9	55.3	56.1	54.5	50.2	47.4	43.7
		MCDB	45.9	45.2	45.2	47.7	51.5	58.4	57.4	58.2	56.5	51.8	49.0	45.3
	10%	WB	43.1	41.9	41.4	43.2	47.7	52.7	54.4	55.1	53.6	48.9	45.8	42.7
		MCDB	44.7	44.3	44.2	46.3	50.4	55.3	56.3	56.9	55.4	50.6	47.7	44.6
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.1	6.1	6.1	5.6	5.9	5.1	5.6	5.8	5.2	5.5	5.1
		MCDBR	6.4	6.9	7.3	8.2	9.7	11.5	8.3	8.8	7.9	6.0	5.6	5.3
		MCWBR	4.9	5.1	5.7	5.4	5.4	6.0	5.3	5.4	5.4	5.1	4.6	4.9
	5% WB	MCDBR	5.4	5.5	6.3	7.2	7.5	10.0	7.6	7.9	6.6	5.0	5.2	5.1
		MCWBR	5.6	5.1	5.5	5.5	5.0	5.8	5.2	5.4	5.1	5.0	5.1	5.3
(40) Clear-Sky Solar Irradiance	taub		0.306	0.315	0.341	0.379	0.376	0.374	0.388	0.372	0.374	0.343	0.311	0.293
	taud		2.322	2.460	2.404	2.297	2.391	2.438	2.439	2.503	2.421	2.408	2.305	2.249
	Ebn at Noon		191	243	262	266	274	276	269	267	247	225	184	161
	Edh at Noon		16	21	28	36	35	33	33	29	28	22	16	14
	RadAvg		136	368	720	1184	1482	1554	1346	1174	776	416	186	103
(45) All-Sky Solar Radiation	RadStd		18	68	91	75	140	141	142	128	106	51	23	17

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	N/A

Nomenclature: See separate page